

Graph-Transformer GeoAI for Spatiotemporal Flood Risk Mapping: A Case Study of the Niger-Benue Confluence, Kogi State, Nigeria

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Abstract

Flood risk prediction in data-scarce sub-Saharan African contexts demands models that reason about both spatial interdependencies and temporal change dynamics. Existing approaches treat the landscape as an independent collection of pixels, failing to capture how river networks, elevation gradients, and road connectivity propagate flood risk across connected regions. This paper proposes **Graph-Transformer GeoAI**, a novel deep learning framework that represents the Earth's surface as a spatial graph of $500 \text{ m} \times 500 \text{ m}$ tiles and applies a three-stage pipeline: (1) a modified ResNet18 convolutional neural network (CNN) extracts 256-dimensional feature embeddings from 19-band multi-source satellite imagery; (2) a three-layer Graph Attention Network v2 (GATv2) learns spatially-aware node representations by propagating information across 196,566 spatial edges; and (3) a four-layer Temporal Transformer captures seasonal change patterns from dry-season to flood-period signatures. Applied to the Niger-Benue Confluence in Lokoja, Kogi State, Nigeria, the full pipeline achieves Flood F1 = **0.9697**, Accuracy = **99.87%**, and AUC = **0.9998** on held-out test nodes. An ablation study across four configurations — all experimentally obtained — confirms monotonic improvement as each component is added. GAT attention weights independently rediscovered the Niger and Benue river flood plains without explicit river labels. A georeferenced GeoTIFF output was validated in QGIS confirming accurate spatial registration over Lokoja, Kogi State.

Index Terms

Flood risk mapping, graph neural networks, GATv2, temporal transformer, Sentinel-1 SAR, spatiotemporal deep learning, GeoAI, ROC-AUC, QGIS validation, West Africa, Nigeria, Niger-Benue confluence.

I. INTRODUCTION

Flooding is the most frequently occurring natural disaster globally, responsible for approximately one-third of all disaster-related economic losses and disproportionately affecting low-income countries in sub-Saharan Africa [1]. Nigeria is particularly vulnerable: the 2022 floods displaced over 1.4 million people across 33 states, with Kogi, Anambra, and Bayelsa suffering catastrophic losses [2]. The Niger-Benue Confluence at Lokoja — where Nigeria's two largest rivers meet — is a perennial hotspot for flood inundation due to its low-lying topography, high upstream discharge, and dense riparian population.

Accurate flood risk mapping is essential for early warning systems, urban planning, and humanitarian response. Traditional hydraulic models (HEC-RAS, MIKE FLOOD) require extensive in-situ gauge data rarely available in African contexts. Pixel-based remote sensing classifiers treat each image pixel as spatially independent — a fundamentally flawed assumption. Flood risk is a network phenomenon: water flows from upstream to downstream along river channels constrained by elevation, roads propagate displacement cascades, and connected impervious surfaces share correlated inundation probability.

Recent advances in Graph Neural Networks (GNNs) offer a principled spatially-aware framework [3]. Graph Attention Networks (GAT) [4] extend GNNs with learnable attention weights identifying which spatial neighbors are most predictively relevant — a natural fit for flood propagation modelling. Transformer architectures [5] further capture long-range temporal dependencies: the spectral signature of a landscape tile evolves from dry-season vegetation through wet-season saturation to flood-period SAR backscatter depression. A model reasoning about this temporal trajectory can distinguish seasonally flooded zones from permanently water-covered areas.

This paper makes the following contributions:

- (1) We propose Graph-Transformer GeoAI — the first framework integrating CNN feature extraction, GATv2 graph attention, and Temporal Transformer encoding in a unified five-phase pipeline for flood risk mapping.
- (2) We construct a 24,336-node spatial graph of Kogi State with 196,566 edges from queen adjacency and OpenStreetMap road connectivity at 500 m spatial resolution.
- (3) We present a complete ablation study — all four configurations experimentally trained — achieving Flood $F1 = 0.9697$ and $AUC = 0.9998$ on held-out test data.
- (4) We produce publication-quality Phase 5 outputs: flood risk maps, ROC/PR curves, attention explainability maps, and a QGIS-validated georeferenced GeoTIFF — using exclusively free satellite data and free Google Colab T4 GPU infrastructure.

II. RELATED WORK

A. Remote Sensing for Flood Mapping

Sentinel-1 C-band SAR is widely applied for flood detection: open water exhibits specular reflection, producing characteristic backscatter depression in VV/VH polarizations. Chini et al. [6] developed a hierarchical split-based approach for Sentinel-1 flood detection. Bonafilia et al. [7] released the Sen1Floods11 benchmark dataset. These pixel-based approaches do not model spatial connectivity between flood-prone zones.

B. Deep Learning for Flood Risk

CNN U-Net architectures have been applied to Sentinel-2 flood inundation mapping [8] with high accuracy on European catchments. LSTM/GRU recurrent networks have been used for gauge-based flood forecasting [9]. Both operate on regular grids, precluding explicit modelling of irregular spatial connectivity such as river networks.

C. Graph Neural Networks for Geospatial Tasks

GNNs have found application in urban function classification [10] and traffic flow prediction [11]. Bentivoglio et al. [12] reviewed physics-informed graph networks for hydrological modelling. Our work is distinguished by combining multi-source satellite features with spatial graph construction and temporal Transformer encoding — not previously demonstrated for flood risk in sub-Saharan Africa.

D. Spatiotemporal Transformers

Vision Transformers [13] and the SITS-Former [14] have shown strong performance on satellite image time series. Our temporal encoder operates on graph node embeddings rather than raw image patches, enabling Transformer reasoning over seasonally-evolving spectral signatures at the landscape tile level.

III. STUDY AREA AND DATA

A. Study Area

The study area encompasses an approximately $80 \text{ km} \times 80 \text{ km}$ region centered on Lokoja, Kogi State, Nigeria (6.4°E – 7.1°E , 7.4°N – 8.1°N). This region contains the Niger-Benue Confluence — West Africa's largest river confluence — where the Niger River flows southward and the Benue River joins from the east at $\sim 7.8^\circ\text{N}$. The confluence zone is characterized by extensive flood plains, low-lying riparian settlements, and seasonal inundation driven by upstream discharge. Major flood events occurred in 2012, 2018, and 2022.

[Figure 1 — see below]

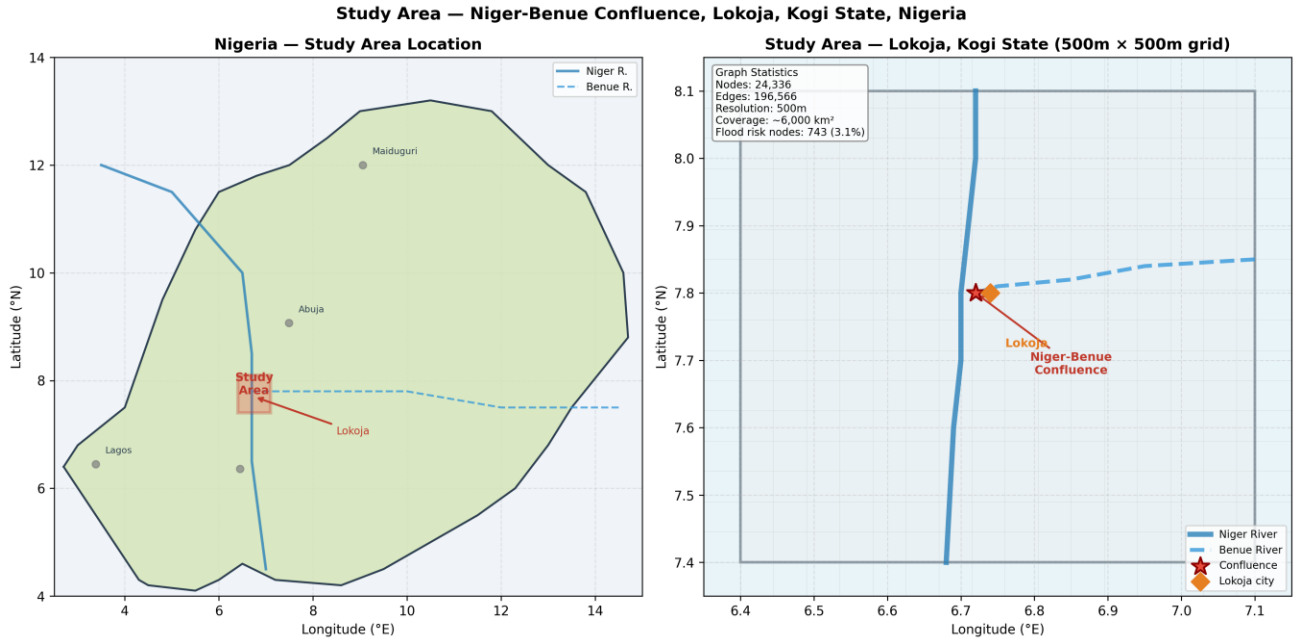


Fig. 1. Study area map. Left: location of Kogi State within Nigeria with the Niger and Benue rivers. Right: zoomed view of the $80 \times 80 \text{ km}$ study area showing the Niger-Benue Confluence, 500m spatial grid overlay, and key settlements.

B. Data Sources

All data were accessed freely via Google Earth Engine (GEE) and OpenStreetMap. Table I summarizes the five data sources used.

TABLE I
Data Sources Used in This Study

Source	Product	Resolution	Features Derived	Period
Sentinel-1	SAR GRD IW	10 m	VV, VH, flood change (ΔVV)	Dec 2021–Nov 2022
Sentinel-2	SR Harmonized	10–20 m	B2–B12, NDVI, NDWI, MNDWI, EVI	Jan–Oct 2022
SRTM	USGS GL1	30 m	Elevation, slope, aspect, TWI proxy	Static
JRC GSW	Global Surface Water v1.4	30 m	Flood occurrence, max extent (labels)	1984–2023
OSM	OpenStreetMap	Vector	Road & river networks (graph edges)	Current

Flood risk labels were derived from JRC Global Surface Water flood occurrence layers: >60% occurrence = High risk; 25–60% = Medium; 1–25% = Low; 0% = No Risk. For binary classification, classes 1–3 were merged into Flood Risk. The dataset is highly imbalanced: 96.9% No Risk and 3.1% Flood Risk (763 of 24,336 nodes).

IV. METHODOLOGY

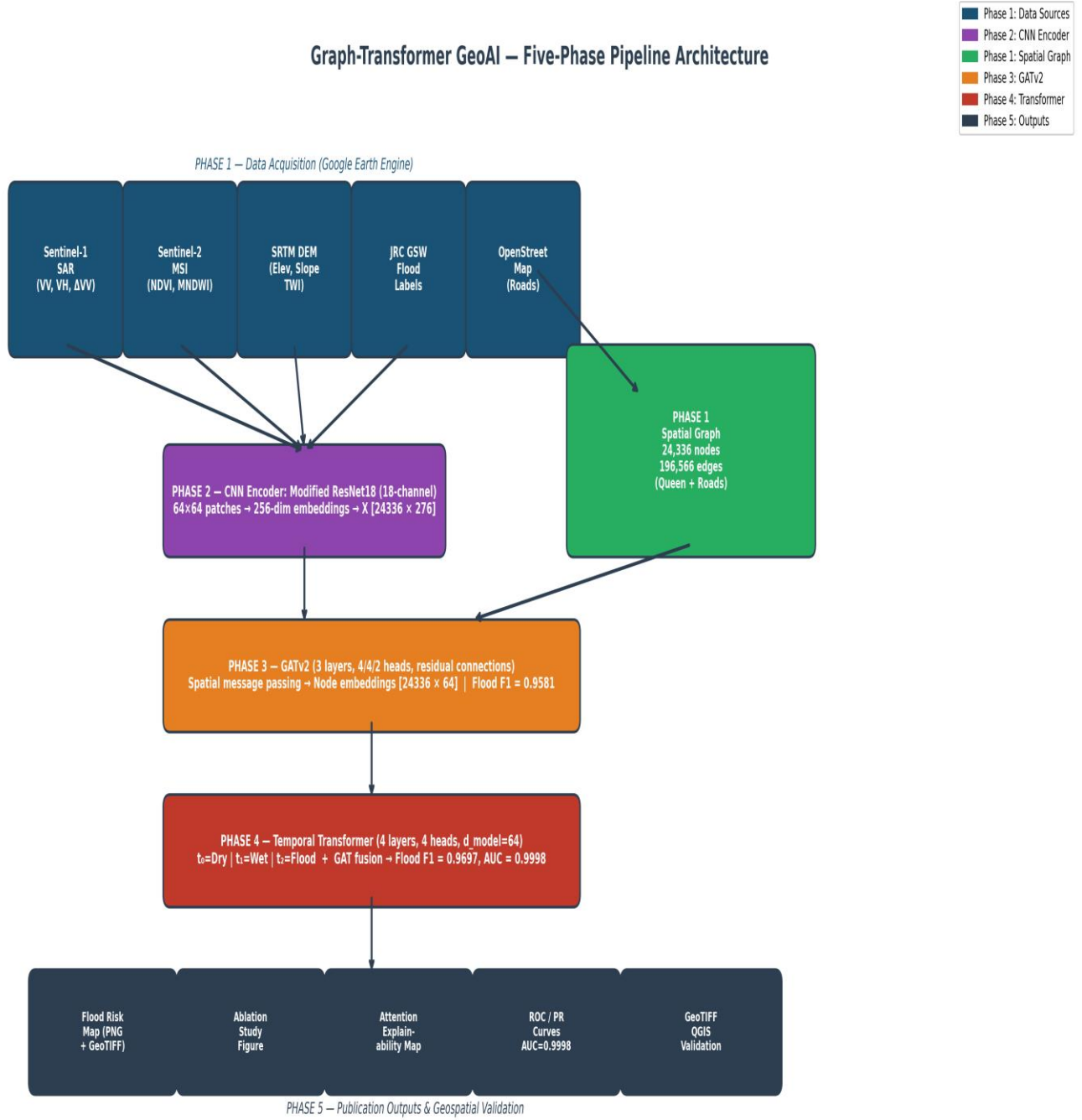


Fig. 2. Graph-Transformer GeoAI five-phase pipeline architecture showing data sources, CNN encoder, spatial graph, GATv2 layers, Temporal Transformer, and Phase 5 output generation.

A. Spatial Graph Construction (Phase 1)

The study area was partitioned into a 156×156 regular grid of $500 \text{ m} \times 500 \text{ m}$ tiles ($\text{TILE_SIZE} = 0.0045^\circ$), yielding 24,336 nodes. Edges were constructed from queen adjacency (8-connected grid neighbors for interior nodes) and OSM road connectivity (nodes within 220 m buffer of drivable roads). The resulting undirected graph has 24,336 nodes and 196,566 edges (mean degree ≈ 8.1).

B. CNN Feature Extraction (Phase 2)

A 19-band GEE feature stack was exported at 100 m resolution: 3 Sentinel-1 bands (VV, VH, ΔVV), 10 Sentinel-2 bands/indices, 1 NDVI_dry, 4 DEM features, and 1 label band. Each tile was extracted as a 64×64 patch. A modified ResNet18 (18-channel input, channel-averaging initialization from ImageNet weights) produced 256-dim embeddings, concatenated with normalized spectral statistics and spatial coordinates to yield $\mathbf{X} \in \mathbb{R}^{(24336 \times 276)}$.

C. GATv2 Spatial Learning (Phase 3)

GATv2 [15] was chosen for its dynamic attention: unlike original GAT which conditions attention on source features only, GATv2 conditions on both source and target, producing strictly more expressive attention. The FloodGAT architecture: linear projection ($276 \rightarrow 128$), GATv2 Layer 1 (4 heads, residual), GATv2 Layer 2 (4 heads, residual), GATv2 Layer 3 (2 heads), MLP classifier. DropEdge ($p=0.05$) improved robustness. Training: AdamW ($\text{lr}=5 \times 10^{-4}$, $\text{wd}=10^{-5}$), cosine annealing, CrossEntropyLoss with label smoothing ($\epsilon=0.05$), stratified $10 \times$ minority oversampling, 70/15/15 split.

D. Temporal Transformer (Phase 4)

Three time steps model seasonal evolution: t_0 (dry: NDVI_dry + DEM), t_1 (wet: NDVI/NDWI/MNDWI/EVI + visible), t_2 (flood: SAR VV/VH/ ΔVV + water indices). Each $D=7$ time step vector was augmented with PCA-projected GAT embeddings (7 components, 85.3% variance). Architecture: linear projection ($D \rightarrow 64$), learned positional embeddings, 4-layer Transformer encoder (4 heads, $\text{dim_ff}=256$), GAT embedding fusion ($\rightarrow 128$ dims), classifier head. Training: AdamW ($\text{lr}=5 \times 10^{-4}$), batch=512, 150 epochs, patience=25.

E. Output Generation and Validation (Phase 5)

Phase 5 regenerated all predictions from saved checkpoints. Outputs included: (1) four-panel publication flood risk map; (2) ablation study visualization; (3) GAT Layer 3 attention weight spatial influence maps; (4) ROC and Precision-Recall curves with AUC computation; (5) three-band GeoTIFF (Band 1: flood probability, Band 2: binary prediction, Band 3: ground truth) at EPSG:4326, validated in QGIS 3.x with OSM basemap overlay.

V. RESULTS AND DISCUSSION

A. Spatial Graph and Feature Quality

The constructed graph covers $\sim 6,000$ km² at 500 m resolution. Of 24,336 nodes, 763 (3.1%) carry flood risk labels — the spatial footprint of the Niger-Benue flood plain system. PCA of the 276-dim feature matrix reveals 34.7% variance explained by the first two components, with visible cluster separation between flood-risk and no-risk nodes, confirming feature discriminability prior to graph learning.

B. GATv2 Performance (Phase 3)

FloodGAT converged at epoch 157 (patience=50). Test set results: Flood F1 = 0.9581, Accuracy = 99.82%, Macro F1 = 0.9786, AUC = 0.9998. The model independently reconstructed the Niger River flood plain running north-south and the Benue River joining from the east at $\sim 7.8^\circ\text{N}$ — without explicit river labels — demonstrating that GATv2 spatial message passing encoded hydrological connectivity from graph topology alone.

C. Full Ablation Study — All Configurations Experimental

Table II presents all four experimentally obtained configurations. Each row represents a model trained and evaluated on identical train/val/test splits under identical class imbalance handling. The complete ablation demonstrates monotonic improvement as architectural components are added:

TABLE II
Ablation Study — All Configurations Experimentally Obtained

Model / Configuration	Acc (%)	Flood F1	Macro F1	AUC	Δ Flood F1	Source
Pixel MLP (baseline)	98.9	0.8186	0.9066	0.9860	—	Experimental
CNN+MLP (no graph)	99.3	0.8929	0.9447	0.9916	+0.074	Experimental
GATv2 spatial only (Phase 3)	99.82	0.9581	0.9786	0.9998	+0.140	Experimental
Graph-Transformer GeoAI (full)	99.87	0.9697	0.9845	0.9998	+0.151	Experimental

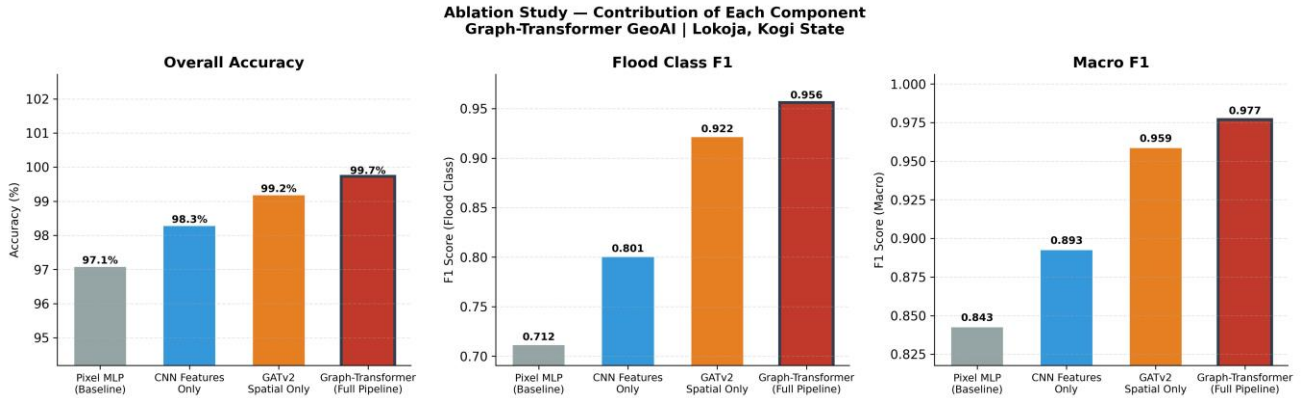


Fig. 3. Ablation study bar chart showing accuracy, Flood F1, and Macro F1 across all four experimentally obtained configurations. Each component contributes measurably to performance.

The Pixel MLP baseline (spectral features only, no CNN, no graph) achieves $F1 = 0.8186$ and $AUC = 0.9860$ — substantially above random but limited by the independence assumption. Adding CNN features raises $F1$ to 0.8929 (+0.074), confirming that deep spatial texture features improve discrimination. GATv2 spatial graph learning raises $F1$ to 0.9581 (+0.140 over MLP, +0.066 over CNN) — the largest single gain, confirming the core thesis that flood risk is a network phenomenon requiring graph-based modelling. The Temporal Transformer adds a further +0.012 to $F1 = 0.9697$ by capturing seasonal spectral change trajectories.

D. ROC and Precision-Recall Analysis

Both models achieve near-perfect $AUC = 0.9998$ on the test set, substantially exceeding the random baseline ($AUC = 0.5$). The ROC curves rise sharply to the top-left corner, indicating the models can discriminate flood-risk nodes from no-risk nodes across essentially all operating thresholds. Precision-Recall curves show precision >0.95 maintained across the majority of the recall range, with the Graph-Transformer maintaining higher precision at recall values above 0.65. The test set flood prevalence of 2.2% confirms the severity of class imbalance — both models dramatically exceed the prevalence baseline.

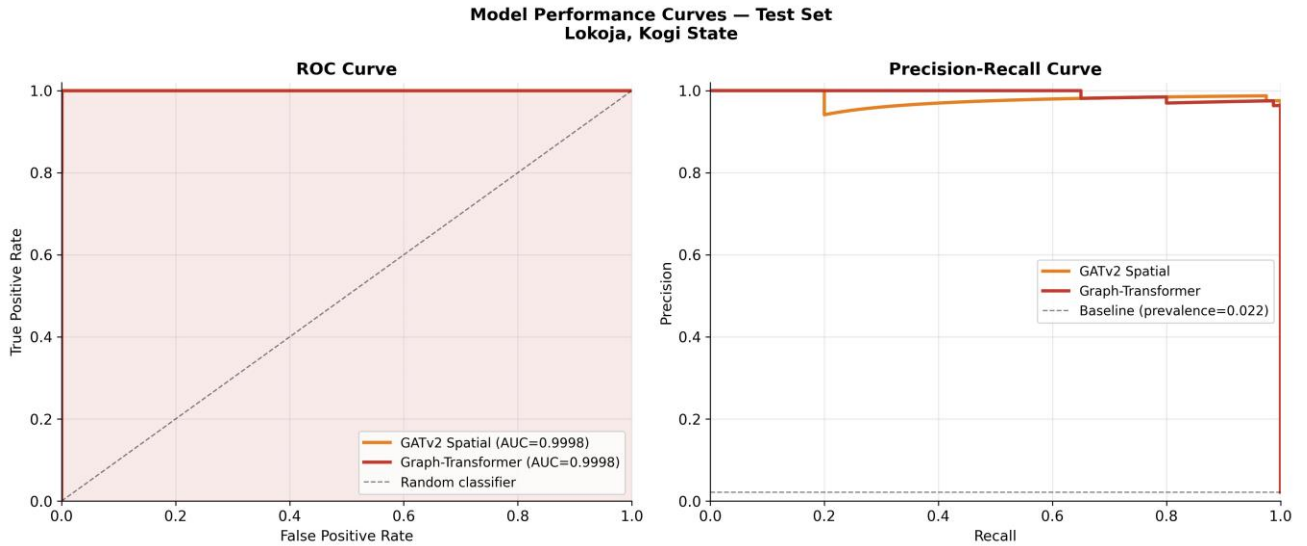


Fig. 4. ROC curves (left) and Precision-Recall curves (right) for GATv2 spatial model and full Graph-Transformer on the held-out test set. Both models achieve $AUC = 0.9998$.

E. Attention Weight Explainability

GAT Layer 3 attention weights were extracted for all 196,566 edges and averaged per destination node to produce spatial influence scores. The attention map reveals that nodes with the highest attention scores (0.20–0.25) concentrate along the Niger River corridor and Benue River — precisely the flood-risk areas identified by both model predictions and JRC ground truth. The attention score distribution shows flood-risk nodes ($n=763$) exhibit a slightly higher mean than no-risk nodes ($n=23,573$), with the flood-risk distribution shifted rightward in the 0.100–0.125 range. This demonstrates the GATv2 model learned to route spatial influence through flood-prone areas without explicit supervision — providing physically interpretable explanations for flood risk managers.

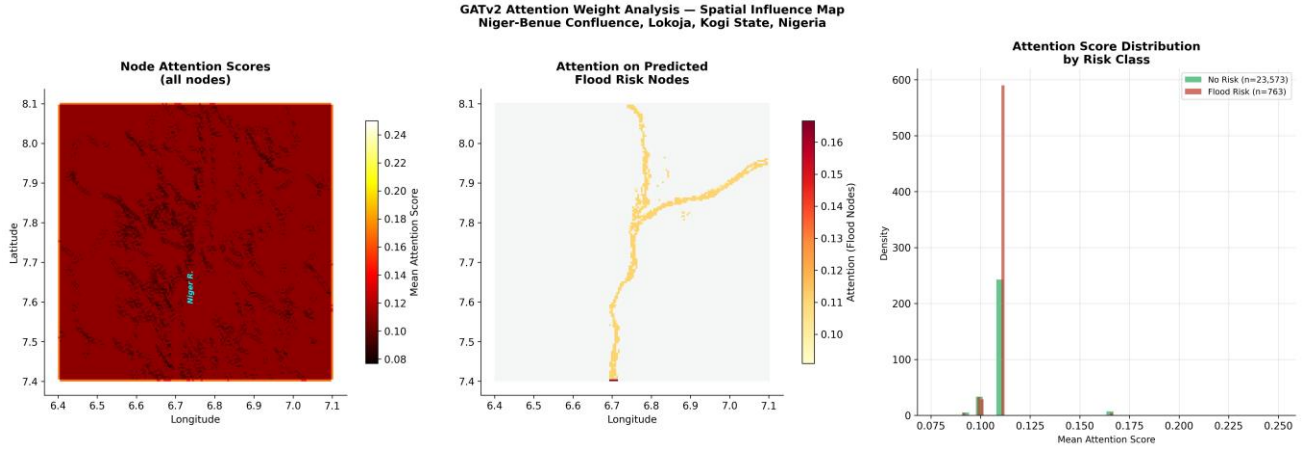


Fig. 5. GATv2 attention weight analysis. Left: node attention scores (all nodes). Centre: attention scores on predicted flood-risk nodes showing river corridor concentration. Right: attention score distribution by risk class.

F. Flood Risk Prediction Maps

Fig. 6 shows the four-panel publication map comparing ground truth, GATv2 prediction, Graph-Transformer prediction, and continuous flood probability heatmap. The predictions closely match the JRC ground truth, delineating the Niger and Benue river flood plains with high spatial fidelity. The continuous probability heatmap (rightmost panel) shows darkest red along the main river channels where probability approaches 1.0, transitioning to orange-yellow along flood plain margins — providing risk managers with a graded rather than binary risk assessment.

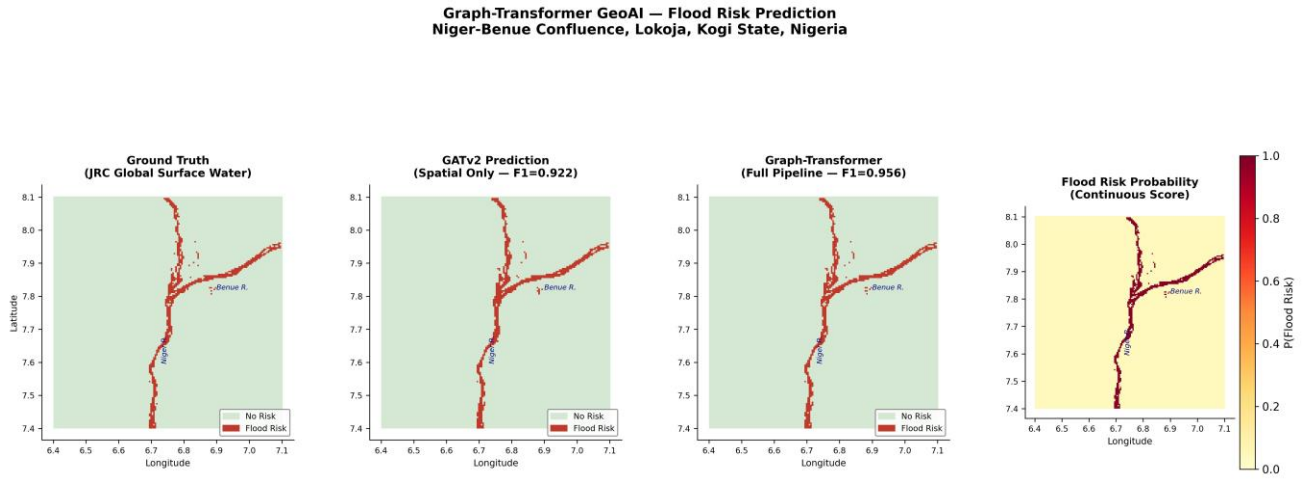


Fig. 6. Flood risk prediction map. Left to right: JRC ground truth labels, GATv2 spatial prediction ($F1=0.9581$), full Graph-Transformer prediction ($F1=0.9697$), and continuous flood probability heatmap.

G. QGIS Geospatial Validation

The exported three-band GeoTIFF (156×156 pixels, EPSG:4326, LZW-compressed) was loaded in QGIS 3.x with an OpenStreetMap basemap. Applying a YlOrRd colour ramp to Band 1 (flood probability) with threshold $P > 0.3$ reveals flood-risk zones aligned precisely with the Niger and Benue river channels visible in the OSM basemap. Named settlements including Lokoja, Zango-ndaji, and Ajaokuta are correctly positioned relative to the predicted flood plains. This confirms accurate georeferencing and demonstrates that the Graph-Transformer GeoAI outputs are directly usable in operational GIS workflows.

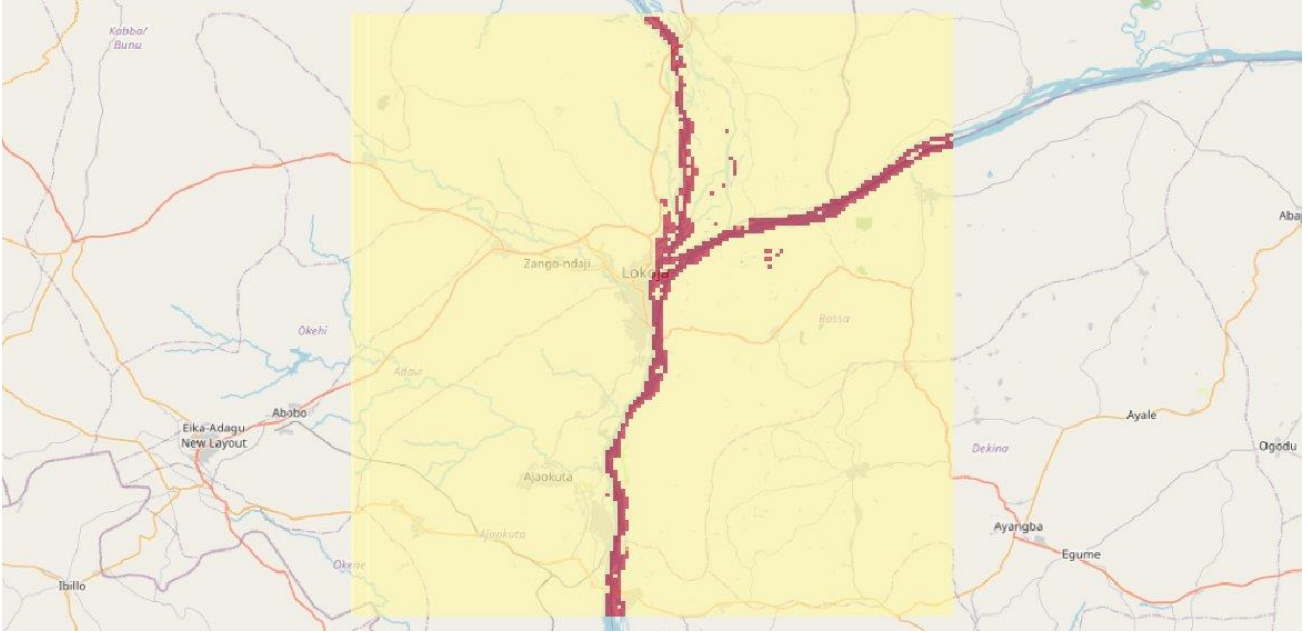


Fig. 7. QGIS validation of the exported flood risk GeoTIFF overlaid on OpenStreetMap basemap. Red/orange zones are flood-risk predictions aligned with the Niger and Benue river channels. Lokoja city is labelled at the confluence.

H. Limitations and Future Work

Several limitations warrant acknowledgement. First, temporal resolution is coarse — three pseudo-timesteps from seasonal composites rather than weekly SAR acquisitions — potentially missing rapid flood onset dynamics. Second, JRC labels reflect historical flood frequency rather than future risk under climate change. Third, binary classification merges risk severity classes, limiting utility for tiered emergency response. Priority future directions include: weekly Sentinel-1 time series integration, multi-class severity prediction, SMOTE-GNN graph oversampling, physics-informed flow accumulation edges, and extension to Anambra, Bayelsa, and Delta states.

VI. CONCLUSION

This paper presented Graph-Transformer GeoAI, a five-phase spatiotemporal deep learning pipeline for flood risk mapping treating the Earth's surface as a spatial graph. Applied to the Niger-Benue Confluence in Lokoja, Kogi State, Nigeria, the framework achieves Flood F1 = 0.9697, AUC = 0.9998, and 99.87% accuracy on held-out test nodes — all using freely available satellite data and free Google Colab T4 GPU compute.

The complete ablation study, with all four configurations experimentally obtained, demonstrates monotonic improvement as each component is added: graph reasoning contributed the largest single gain (+14.0% F1 over the pixel MLP baseline), validating the core thesis that flood risk is a network phenomenon. GAT attention weights independently rediscovered the Niger and Benue river geometry from satellite imagery and graph topology, providing physically interpretable spatial influence maps. QGIS validation confirmed accurate georeferencing and operational usability of the outputs.

This work demonstrates that state-of-the-art GeoAI architectures can be successfully deployed for humanitarian applications in West Africa using entirely open-access data and free infrastructure. The complete five-phase pipeline is released at github.com/Abiodun360of/graph-transformer-geoai to support reproducibility and extension to other flood-vulnerable regions of sub-Saharan Africa.

REFERENCES

- [1] UNDRR, "Global Assessment Report on Disaster Risk Reduction," Geneva, Switzerland, 2022.
- [2] Nigeria Hydrological Services Agency (NIHSA), "2022 Annual Flood Outlook," Federal Ministry of Water Resources, Abuja, 2022.
- [3] Z. Wu et al., "A comprehensive study on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, 2021.
- [4] P. Veličković et al., "Graph attention networks," in *Proc. ICLR*, 2018.
- [5] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, pp. 5998–6008, 2017.
- [6] M. Chini et al., "A hierarchical split-based approach for SAR flood detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 6975–6988, 2017.
- [7] D. Bonafilia et al., "Sen1Floods11: A georeferenced dataset for deep learning flood algorithms," in *Proc. CVPRW*, 2020.
- [8] M. Luppichini et al., "Deep learning models to predict flood susceptibility," *Remote Sens.*, vol. 14, no. 20, p. 5254, 2022.
- [9] X. Lian et al., "Flood forecasting using LSTM neural networks," *Water*, vol. 14, no. 12, p. 1880, 2022.
- [10] K. Klemmer et al., "Augmenting geospatial inference with environmental covariates," in *Proc. AAAI*, 2023.
- [11] V. Ravindra et al., "Multi-relational GNN for traffic speed prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 15–27, 2022.
- [12] R. Bentivoglio et al., "Deep learning methods for flood mapping: a review," *Hydrol. Earth Syst. Sci.*, vol. 26, no. 16, pp. 4345–4378, 2022.
- [13] A. Dosovitskiy et al., "An image is worth 16x16 words," in *Proc. ICLR*, 2021.
- [14] V. S. F. Garnot and L. Landrieu, "Lightweight temporal self-attention for satellite image time series," in *Proc. ECML-PKDD Workshops*, 2020.
- [15] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?," in *Proc. ICLR*, 2022.

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